

Отчёт к лабораторной работе 5

Тема – «Партицирование в PostgreSQL»

Добавим в выбранную таблицу данные для соблюдения условий:

```
INSERT INTO booking (booking_id, flight_id, booking_date) VALUES
(6, 1, '2024-02-01 10:00:00'),
(7, 2, '2024-02-15 11:00:00'),
(8, 3, '2024-02-20 12:00:00'),
(9, 4, '2024-02-25 13:00:00'),
(10, 5, '2024-02-28 14:00:00'),
(11, 1, '2024-01-05 10:00:00'),
(12, 2, '2024-01-10 11:00:00'),
(13, 3, '2024-01-15 12:00:00'),
(14, 4, '2024-01-20 13:00:00'),
(15, 5, '2024-01-25 14:00:00');
```

Создадим таблицы-партиции:

```
CREATE TABLE booking_2024_03 (
    CHECK (booking_date >= '2024-03-01' AND booking_date < '2024-04-01')
) INHERITS (booking);
```

```
CREATE TABLE booking_2024_02 (
    CHECK (booking_date >= '2024-02-01' AND booking_date < '2024-03-01')
) INHERITS (booking);
```

```
CREATE TABLE booking_2024_01 (
    CHECK (booking_date >= '2024-01-01' AND booking_date < '2024-02-01')
) INHERITS (booking);
```

Создадим функцию, обеспечивающую партицирование:

```

CREATE OR REPLACE FUNCTION booking_insert_trigger()
RETURNS TRIGGER AS $$
BEGIN
    IF (NEW.booking_date >= '2024-03-01' AND NEW.booking_date < '2024-04-01')
    THEN
        INSERT INTO public.booking_2024_03 VALUES (NEW.*);
    ELSIF (NEW.booking_date >= '2024-02-01' AND NEW.booking_date < '2024-03-01')
    THEN
        INSERT INTO public.booking_2024_02 VALUES (NEW.*);
    ELSIF (NEW.booking_date >= '2024-01-01' AND NEW.booking_date < '2024-02-01')
    THEN
        INSERT INTO public.booking_2024_01 VALUES (NEW.*);
    ELSE
        RAISE EXCEPTION 'Дата бронирования вне диапазона партиций.';
    END IF;
    RETURN NULL;
END;
$$ LANGUAGE plpgsql;

```

Подключим функцию к мастер-таблице:

```

CREATE TRIGGER booking_insert_trigger
BEFORE INSERT ON booking
FOR EACH ROW EXECUTE FUNCTION booking_insert_trigger();

```

Перенесём данные из мастер-таблицы в партиции:

```

INSERT INTO booking_2024_03
SELECT * FROM booking
WHERE booking_date >= '2024-03-01' AND booking_date < '2024-04-01';

INSERT INTO booking_2024_02
SELECT * FROM booking

```

```
WHERE booking_date >= '2024-02-01' AND booking_date < '2024-03-01';
```

```
INSERT INTO booking_2024_01
```

```
SELECT * FROM booking
```

```
WHERE booking_date >= '2024-01-01' AND booking_date < '2024-02-01';
```

Очистим мастер-таблицу:

```
DELETE FROM ONLY booking
```

```
WHERE booking_date >= '2024-01-01' AND booking_date < '2024-04-01';
```

Добавим в неё новые данные:

```
INSERT INTO booking (booking_id, flight_id, booking_date) VALUES
```

```
(16, 1, '2024-03-04 15:00:00'),
```

```
(17, 5, '2024-02-14 10:00:00'),
```

```
(18, 5, '2024-01-01 10:00:00');
```

Проверим сообщение об ошибке при вводе некорректных данных:

```
INSERT INTO booking (booking_id, flight_id, booking_date) VALUES
```

```
(6, 1, '2024-04-01 10:00:00');
```

```
ERROR: Дата бронирования вне диапазона партиций.
```

```
CONTEXT: PL/pgSQL function booking_insert_trigger() line 19 at RAISE
```

Проверка таблиц:

```
postgres=# select * from booking_2024_03;
```

booking_id	flight_id	booking_date
1	3	2024-03-17 16:04:14
2	3	2024-03-14 17:35:03
3	2	2024-03-15 11:15:35
4	4	2024-03-12 14:27:56
5	1	2024-03-16 15:09:14
16	1	2024-03-04 15:00:00

```
(6 rows)
```

```
postgres=# select * from booking_2024_02;
 booking_id | flight_id | booking_date
-----+-----+-----
          6 |          1 | 2024-02-01 10:00:00
          7 |          2 | 2024-02-15 11:00:00
          8 |          3 | 2024-02-20 12:00:00
          9 |          4 | 2024-02-25 13:00:00
         10 |          5 | 2024-02-28 14:00:00
         17 |          5 | 2024-02-14 10:00:00
(6 rows)
```

```
postgres=# select * from booking_2024_01;
 booking_id | flight_id | booking_date
-----+-----+-----
         11 |          1 | 2024-01-05 10:00:00
         12 |          2 | 2024-01-10 11:00:00
         13 |          3 | 2024-01-15 12:00:00
         14 |          4 | 2024-01-20 13:00:00
         15 |          5 | 2024-01-25 14:00:00
         18 |          5 | 2024-01-01 10:00:00
(6 rows)
```

```
postgres=# select * from booking;
 booking_id | flight_id | booking_date
-----+-----+-----
          1 |          3 | 2024-03-17 16:04:14
          2 |          3 | 2024-03-14 17:35:03
          3 |          2 | 2024-03-15 11:15:35
          4 |          4 | 2024-03-12 14:27:56
          5 |          1 | 2024-03-16 15:09:14
         16 |          1 | 2024-03-04 15:00:00
          6 |          1 | 2024-02-01 10:00:00
          7 |          2 | 2024-02-15 11:00:00
          8 |          3 | 2024-02-20 12:00:00
          9 |          4 | 2024-02-25 13:00:00
         10 |          5 | 2024-02-28 14:00:00
         17 |          5 | 2024-02-14 10:00:00
         11 |          1 | 2024-01-05 10:00:00
         12 |          2 | 2024-01-10 11:00:00
         13 |          3 | 2024-01-15 12:00:00
         14 |          4 | 2024-01-20 13:00:00
         15 |          5 | 2024-01-25 14:00:00
         18 |          5 | 2024-01-01 10:00:00
(18 rows)
```

```
postgres=# select * from only booking;
 booking_id | flight_id | booking_date
-----+-----+-----
(0 rows)
```